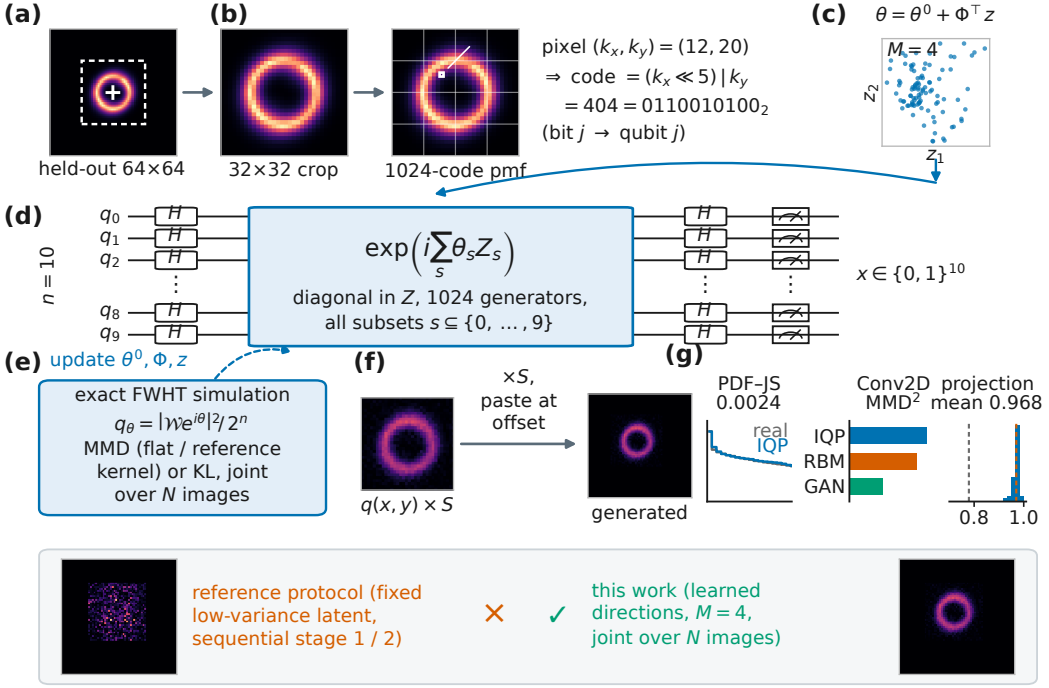

(a)
held-out 64×64
(b)
32×32 crop
1024-code pmf
pixel $(k_x, k_y) = (12, 20)$
$\Rightarrow$ code $= (k_x \ll 5) \,|\, k_y$
$= 404 = 0110010100_2$
(bit j $\rightarrow$ qubit j)
(c)
$\theta = \theta^0 + \Phi^\top z$
$M = 4$
z_2
z_1
(d)
$n = 10$
q_0
q_1
q_2
q_8
q_9
H
$\exp\left(i \sum_s \theta_s Z_s\right)$
diagonal in Z, 1024 generators,
all subsets $s \subseteq \{0, \ldots, 9\}$
$x \in \{0, 1\}^{10}$
(e)
update θ^0, Φ, z
exact FWHT simulation
$q_\theta = |\mathcal{W} e^{i\theta}|^2 / 2^n$
MMD (flat / reference
kernel) or KL, joint
over N images
(f)
$q(x, y) \times S$
$\times S$,
paste at
offset
generated
(g)
PDF–JS
0.0024
real
IQP
Conv2D projection
MMD^2 mean 0.968
IQP
RBM
GAN
0.8
1.0
reference protocol (fixed
low-variance latent,
sequential stage 1 / 2)
this work (learned
directions, $M = 4$,
joint over N images)